

Expanding the dialogue: A closer look at AIH management and HCC risk

To the Editor:

I am writing to share my reflections on the recent publication in your prestigious journal titled "Incidence and predictors of hepatocellular carcinoma in patients with autoimmune hepatitis",¹ which discussed autoimmune hepatitis (AIH) and its progression to hepatocellular carcinoma (HCC). This study notably bridges a significant knowledge gap in our understanding of the progression from AIH to HCC, leveraging the largest multicentric AIH cohort to date. This letter aims to contribute constructive perspectives that might enrich the authors' findings for the readership, while also encouraging further exploration in this pivotal domain.

Firstly, Colapietro *et al.* observed in their study that individuals who did not respond to immunosuppressive therapy, predominantly prednisolone, faced an elevated risk of cirrhosis, yet no discernible difference in HCC development between responders and non-responders was reported.¹ This unexpected result was attributed to the limited number of tumours in their cohort.¹ In fact, the duration of immunosuppressive treatment might play a role, as highlighted by previous research. Specifically, the synergistic effect of oral corticosteroid use alongside traditional risk factors for HCC (such as alcohol-related disease, chronic liver disease, and diabetes mellitus) has been documented, with an adjusted odds ratio (OR) of 33.7.² Lai *et al.* also demonstrated a duration-dependent relationship between oral corticosteroid use and HCC risk, showcasing an adjusted OR of 1.03 for every additional year of corticosteroid use among patients with any of the mentioned comorbidities, compared to non-use; this OR increases to 1.11 among patients without these comorbidities.² Moreover, Oh *et al.* found that the risk of liver cancer was 1.46-fold higher among individuals prescribed oral glucocorticoids for ≥ 30 days continuously compared to those not or minimally exposed to glucocorticoids.³ While these studies did not exclusively focus on patients with AIH, there is no reason to exclude similar effects of long-term corticosteroids use on HCC risk among patients with AIH. Thus, I encourage the original authors to review patients' immunosuppressive therapy records and perform subgroup analyses to elucidate the relationship between treatment duration and HCC risk. Identifying a link between cumulative duration of corticosteroid use and HCC in patients with AIH could significantly impact their clinical management, especially considering the potential necessity for long-term treatment in managing AIH and concurrent HCC risk factors.²

Secondly, although AIH is relatively uncommon,⁴ from a clinical standpoint, a patient with AIH essentially faces two outcomes: developing HCC (100% probability) or not (0%

probability). The original authors argued that due to the very low HCC risk among patients with AIH without other advanced liver disease, routine surveillance for this subgroup may not be justified.¹ However, from a patient-centered view, it seems prudent to focus attention on those at greater HCC risk, while advocating for lifestyle modifications (e.g., minimizing obesity, healthy eating, avoiding tobacco and alcohol, increasing physical activity) that could mitigate modifiable HCC risks without inducing potential harm. Such approaches, along with promoting hepatitis B vaccination, might be more beneficial than routine HCC surveillance alone,⁵ as they could minimize the already low HCC risk among patients with AIH, eliminating the need for frequent monitoring and thus reducing both psychological stress and the likelihood of false-positive results during routine screening.⁶

Lastly, given the rarity of HCC among patients with AIH and the logistical and costly challenges of long-term prospective clinical follow-up, considering the cost-effectiveness of continuous routine HCC surveillance, the psychological impact on patients, and the risk of false-positive results,^{4,6} leveraging deep-learning-based risk modelling to predict AIH progression to HCC represents a promising avenue.⁷ This innovative approach, albeit not yet widely adopted in clinical practice, could offer valuable insights into disease patterns and progression risks, marking a crucial area for future research. For instance, Liang *et al.* have underscored the potential of a deep learning model to predict HCC risk a year in advance using minimal features (entirely based on patients' age, gender, diseases, and drug data) from electronic health records, achieving high sensitivity and specificity in prediction outcomes.⁸

Dan Shan^{1,2,*}

¹Department of Biobehavioural Sciences, Columbia University, New York, USA

²Faculty of Health and Medicine, Lancaster University, Lancaster, UK

*Corresponding author. Address: Department of Biobehavioural Sciences, Columbia University, Building 528, 525 W 120th St Suite 1159, New York, NY 10027, United States; Tel.: (212) 555-1229.

E-mail address: ds3806@tc.columbia.edu

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Conflict of interest

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Supplementary data

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